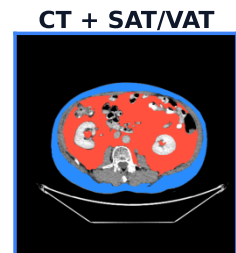


Systematic study of data-scarce body-composition segmentation

TWO PRIMARY DATASETS

FOUR EXPERIMENTAL FACTORS

EVALUATION



AATTCT-IDS (CT)

300 subjects · 3213 slices
512×512 · SAT/VAT (3 cls)



Thigh Dixon MRI

25 subjects · 1513 slices
256×256 · 4ch · 14 cls

Subject-level split 70 / 10 / 20

no slice leakage across sets

Factor 1 . Architecture

U-Net | U-Net++ | DeepLabV3+ | SegFormer

Factor 2 . Pretraining source

random | ImageNet | CT | self-supervised (SSL)

Factor 3 . Input design

+fat-fraction (primary) · +SE gate (ablation)

Factor 4 . Multi-task learning

shared encoder, two task decoders

Per-subject Dice & IoU

held-out subjects

Data-efficiency + 5-fold CV

N=2,4,8,18; per-subject

3 seeds -> mean +/- std

channel-weight & occlusion

Findings: (1) architecture differences comparatively small (2) ImageNet and CT initialisation statistically equivalent
(3) fat-fraction proxy: significant gain at the lowest labels (4) shared encoder close to single-task